aRES_ppi — project overview

How does aRES recognise phage proteins, and what activates its NADase?

Rotation project

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This document is the single-page-in, many-pages-out view of the project: what is established, what rests on a prediction, and what is simply missing. The missing parts are numbered **G1–G13** and collected in the register below — write into the boxes and the register updates on the next build.

Source material is the markdown in `docs/01–docs/08` and the Obsidian vault in `notes/vault/`; this file does not replace them, it marks where they stop.

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1 The question, and the claim

The system. aRES is an anti-phage defence protein: an N-terminal domain of unknown function (`PF18870`, "HEPN/RES N-terminal domain 1", NTD-1) fused to a C-terminal RES NADase (`PF08808`). Osterman et al. 2026 (*Cell Host & Microbe* 34:991–999) establish that aRES is activated by direct binding to T7 DNA polymerase (gp5), and explicitly leave the recognition mechanism open.

The claim this project tests. NTD-1 is a *modular phage-protein sensor*: it binds the trigger protein and thereby activates the RES NADase. The modular version matters more than the minimal one — it predicts that swapping NTD-1 between aRES homologues swaps the phage range.

Why NTD-1 and not something else. There is no experimental structure of any PF18870 domain anywhere. That is simultaneously the project's risk and its cheapest novelty.

2 What is established, by provenance

The single most useful column here is the last one. Phage genetics survives whatever happens to the models; the models do not survive contact with the genetics.

Statement	Strength	Where it comes from
aRES is activated by direct binding to T7 gp5	Established	Paper, Fig. 4
gp5 **S122P** and **D106G** each escape *both* the UC5 and S_34AVR aRES	Established	Phage genetics — two independent isolates; the hardest evidence in the project
The gp5 epitope is therefore the ∼100–125 loop	Strong	Follows from the genetics alone, not from any model
T7 gp5 is a heterodimer with host thioredoxin; the activating species in vitro was the **gp5·TrxA holoenzyme** (NEB M0274S)	Established	Paper's methods — so docking gp5 alone may be docking the wrong species
λ-vir has no DNA polymerase, yet aRES defends against it	Established	λ replicates with host Pol III; its only replication proteins are O and P
Therefore **at least one aRES trigger is not a polymerase**	Strong	Direct consequence of the row above — the paper's framing holds for T7, not as a general rule
aRES carries six Cys in residues 1–120 (C8, C11, C31, C34, **C78, C96**)	Verified	Re-checked against the repository's own coordinates
No lysine anywhere in aRES 49–120	Verified	Same — all ten N-block lysines sit in the Zn ribbon. This constrains amine-reactive crosslinking directly
gp5 Ser122 OG hydrogen-bonds aRES Asp104 (3.06 Å)	**Weak**	One AlphaFold3 cofold, one seed, poor ipTM. *Not* load-bearing — see §3
aRES is a monomer, and the complex is 1:1	**Assumed**	Never measured. The only solved RES-NADase structure (6GW6) is built on a RES toxin *dimer*

3 The five predictions the project rests on

Each of these is currently a prediction, and each has a cheap experiment that kills it. **P1 and P5 are free and belong in week 1** — they are the two cheapest ways to find out that a downstream month was going to be wasted.

P1 — the AlphaFold3 cofold interface is real. Current evidence: one cofold, one pose. The killing experiment is an interface-confidence audit: ipTM, interface PAE, and the per-seed frequency of the D104–S122 contact. If it appears in 1 of 5 seeds, stop before ordering a primer.

G1 — P1 interface-confidence audit

Fill in from the cofold outputs: ipTM = _____1.6cm, mean interface PAE = _____1.6cm Å, D104–S122 contact present in _____0.8cm of _____0.8cm seeds. Blocked only by GG11 (the model files are not in the repo). This is free and gates everything downstream of the cofold.

P2 — the gp5 100–125 loop contacts the aRES N-block. Note the restatement: the earlier form, "Ser122 contacts Asp104", overstates one low-ipTM seed. Killed by zero-length XL-MS (gp5 K103/K114/K118 to an aRES acidic residue in 49–120), and by the **gp5(Δ118–123)·TrxA** activation test — a published, catalytically competent deletion (1T7P/1TK0) that removes G121/S122 outright. If Δ118–123 still activates aRES, S122 is not the contact.

G2 — Does gp5(Δ118–123) still activate aRES?

One clone, one activation assay — and it decides whether the entire S122 story survives. Result: _____3cm. This is the highest information-per-effort experiment in the project.

P3 — NTD-1 is the sensor module. Evidence is the domain architecture plus the authors' own statement; notably *not* the cofold. Killed by asking whether the isolated N-block binds gp5·TrxA at all (MST / nanoDSF), and whether excess N-block *competes in trans* against gp5-driven activation of full-length aRES.

P4 — aRES carries a structural Zn(Cys)$_4$ site. Evidence so far is AlphaFold SG–SG geometry only (3.55–3.81 Å against an ideal 3.76; no metal in the model). Killed or confirmed by ICP-MS stoichiometry, native ESI-MS (+63.4 Da), and Zn K-edge XANES.

G3 — Zn stoichiometry

Zn per aRES monomer by ICP-MS/OES: _____1.5cm. Native ESI-MS mass shift: _____1.5cm Da. Coordination number and ligand type (S vs N/O) from XANES: _____3cm.

P5 — aRES is a monomer and the complex is 1:1. Assumed throughout, measured never. One SEC-MALS or AUC run settles it. If aRES dimerises, apo aRES is ∼100 kDa — unresolvable from 92 kDa gp5·TrxA by mass photometry — and every mass, mask volume and particle count in the cryo-EM plan is wrong.

G4 — Oligomeric state of apo aRES

SEC-MALS or AUC, one run, before any stoichiometry is quoted anywhere: measured mass = _____1.8cm kDa, so aRES is a _____2.2cm.

4 Numbers that do not exist yet

Every entry below is currently blank in the project, and each one changes a downstream decision rather than merely adding detail.

G5 — The dissociation constant

K_d for aRES ↔ gp5·TrxA = _____2cm. Nobody has measured it. Every technique in the interaction track has a concentration window, so this number decides which methods are even applicable — and a K_d above ∼50 µM takes the interaction track off the table.

G6 — Does the isolated N-block express solubly, fold, and load zinc?

Yes / no, plus yield per litre: _____2cm. This single answer gates the whole domain track — the first experimental structure of any PF18870 domain.

G7 — Does recombinant gp5·TrxA exist at ≥0.5 mg by week 6?

Yes / no: _____1.2cm. Gates the whole interaction track. The commercial NEB enzyme cannot supply this — it has to be co-expressed.

The scheduling consequence. The N-block expression screen and the gp5·TrxA co-expression must start in the *same* week, in parallel. The fallback is directional: if the interaction work collapses, the domain track continues untouched, but if the N-block is insoluble the domain track dies while the interaction track continues on full-length aRES. Serialising the two is the single most likely way this ends with protein in a freezer and no data.

5 Open contradictions

These are not missing measurements — they are places where two sources disagree, and each one sits upstream of something expensive.

The λ problem. λ-vir encodes no DNA polymerase, yet two of the three aRES proteins defend against it. So the polymerase model is incomplete. The candidate resolution is that NTD-1 senses phage proteins that *hijack host replication factors*: T7 gp5 captures thioredoxin, λ P captures DnaB. One co-expression screen tests it.

G8 — Which λ protein triggers aRES?

Co-express λ P (then O, then Gam) with UC5 aRES in the Fig. 4G format, S_34AVR as the negative control. Trigger identified: _____2.5cm. A positive result here reframes the project from "polymerase sensor" to "replication-hijack sensor".

The 234/293 mismatch. The model has **Ile234** and **Ser293**; the paper's defence-null list says **S234A** and **V293A**. The pattern — a Ser at 293 and a branched aliphatic at 234 — is what a transposed pair of residue numbers looks like. Four of six positions verify, so this is localised rather than a frame error, but it is upstream of every mutant in the plan.

G9 — Resolve 234/293 against the primary sequence record

Is it a transposition in the paper (i.e. S293A and V234A), or a different UC5 isolate? Answer: _____3cm. Resolve before ordering anything.

The RES Pfam accession. The paper is internally inconsistent: Fig. 1B labels the domain `PF08808 RES` while the STAR Methods text says otherwise. An HMM search built on the wrong accession returns a histone-like family and would silently corrupt the homolog collection for the conservation analysis.

G10 — Confirm the RES accession on InterPro

Accession in use: _____2cm. Confirm before building the HMM — the failure mode here is silent, not loud.

6 Data and files still missing

G11 — The cofold model files

The vault has figures but no `.cif`/`.pdb`. With the coordinates in `data/structures/`, `analysis/scripts/interface_contacts.py` gives the full interface contact list, the NTD-1 vs RES breakdown, and the seed-to-seed reproducibility of the S122–D104 contact in one command — which is exactly G1. Status: _____2cm.

G12 — Supplementary Tables S2/S3/S4 and Fig. 1B at full resolution

Needed for the homolog set and to finish decoding the phage-range matrix. Status: _____2cm.

G13 — The blind HADDOCK run output

In `analysis/docking/blind_run/`. Needed to compare blind against the one-sided restrained run, where both escape residues turned out to be buried. Status: _____2cm.

7 What the gaps decide

Gap	What it unblocks or kills
G1, GG11	Whether the cofold interface deserves any further resource at all
G2	Whether S122 is the contact — one clone decides it
G3, G6	The domain track: Zn validation and the first PF18870 structure

Gap	What it unblocks or kills
G4	Which biophysical technique is applicable; whether the interaction track is viable
G5	Every mass, mask volume and particle count quoted downstream
G8	Whether the project is about polymerases or about replication hijack
G9, G10	Whether the mutants and the homolog set are built on correct identifiers

Realistic outcome. A rotation cannot deliver a structure of the 143 kDa aRES·gp5·TrxA complex. It can plausibly deliver one of two publishable things — the first experimental structure of a PF18870 domain (∼30–35%), or a measured K_d with a residue-level interface restraint set (∼40–45%) — with roughly a 60% chance of at least one. The plan keeps both live until a decision point kills one.